

## LITSCHAU, Austria

WMO: 110210

Lat: 48.955N

Lon: 15.038E

Elev: 1826

StdP: 13.75

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 6.3        | 12.7 | 1.8                           | 6.4 | 7.9  | 7.8 | 8.8 | 14.2 | 11.8                  | 32.9 | 10.5 | 29.8 | 1.8                   | 110  | 0.497 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.0                   | 84.5            | 66.9 | 80.9 | 65.4 | 77.6 | 63.8 | 68.4                | 80.3 | 66.6 | 77.8 | 65.1 | 75.1 | 3.7                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.6                            | 97.6 | 72.6 | 62.9 | 91.7 | 70.5 | 61.3 | 86.7 | 69.3 | 33.7          | 80.2 | 32.3 | 78.2 | 31.0 | 75.4 | 74.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.8           | 8.7  | 7.5 | -0.9                       | 90.3 | 7.0                | 3.5 | -6.0                                               | 92.9 | -10.1      | 94.9 | -14.0      | 96.9 | -19.1      | 99.4 |    |    |
| (5)               |      |     | -1.4                       | 71.3 | 6.8                | 2.0 | -6.2                                               | 72.8 | -10.2      | 73.9 | -14.0      | 75.0 | -19.0      | 76.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 46.3   | 27.8  | 30.5  | 37.1  | 46.3  | 54.5  | 60.6  | 63.9  | 63.6  | 54.9  | 46.5  | 37.7  | 30.6  |  |  |
|                                                                        | DBStd       | 14.70  | 8.57  | 8.69  | 7.54  | 7.86  | 6.83  | 6.87  | 6.06  | 6.27  | 6.46  | 7.03  | 6.97  | 7.98  |  |  |
|                                                                        | HDD50       | 2982   | 689   | 546   | 402   | 159   | 33    | 3     | 0     | 1     | 24    | 153   | 372   | 600   |  |  |
|                                                                        | HDD65       | 7005   | 1154  | 966   | 865   | 561   | 331   | 165   | 96    | 101   | 309   | 572   | 820   | 1065  |  |  |
|                                                                        | CDD50       | 1617   | 0     | 0     | 2     | 48    | 172   | 323   | 430   | 424   | 171   | 45    | 2     | 0     |  |  |
|                                                                        | CDD65       | 164    | 0     | 0     | 0     | 0     | 5     | 34    | 61    | 59    | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1635   | 0     | 0     | 0     | 8     | 50    | 344   | 599   | 582   | 52    | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 405    | 0     | 0     | 0     | 0     | 4     | 74    | 160   | 161   | 6     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 3.2    | 3.6   | 3.6   | 3.7   | 3.5   | 3.3   | 3.0   | 2.9   | 2.6   | 2.7   | 3.1   | 3.4   | 3.4   |  |  |
|                                                                        | PrecAvg     | 27.6   | 1.5   | 1.3   | 1.9   | 1.7   | 3.1   | 3.6   | 3.7   | 3.4   | 2.4   | 1.8   | 1.7   | 1.5   |  |  |
| (15) Precipitation                                                     | PrecMax     | 40.9   | 3.5   | 3.1   | 3.9   | 3.2   | 6.9   | 7.4   | 7.3   | 12.7  | 6.5   | 4.8   | 3.1   | 3.0   |  |  |
|                                                                        | PrecMin     | 20.9   | 0.2   | 0.2   | 0.6   | 0.1   | 0.7   | 1.3   | 1.2   | 0.9   | 0.3   | 0.3   | 0.0   | 0.2   |  |  |
|                                                                        | PrecStd     | 4.4    | 0.8   | 0.8   | 0.9   | 0.9   | 1.3   | 1.5   | 1.3   | 2.3   | 1.3   | 1.1   | 0.8   | 0.6   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 51.1  | 55.1  | 63.2  | 74.1  | 79.0  | 86.2  | 89.4  | 89.9  | 79.9  | 69.2  | 57.8  | 50.3  |  |  |
|                                                                        |             | MCWB   | 44.5  | 44.1  | 50.0  | 57.7  | 62.5  | 67.4  | 67.8  | 68.0  | 64.2  | 57.7  | 51.6  | 45.9  |  |  |
|                                                                        | 2%          | DB     | 46.3  | 49.5  | 57.6  | 68.5  | 74.8  | 81.8  | 84.3  | 84.4  | 74.4  | 63.8  | 53.4  | 46.5  |  |  |
|                                                                        |             | MCWB   | 41.7  | 43.0  | 47.3  | 54.9  | 61.0  | 66.1  | 66.8  | 66.7  | 61.9  | 56.0  | 49.2  | 43.4  |  |  |
|                                                                        | 5%          | DB     | 42.9  | 45.7  | 53.0  | 64.2  | 71.4  | 77.9  | 80.4  | 80.4  | 70.0  | 60.1  | 50.5  | 43.7  |  |  |
|                                                                        |             | MCWB   | 39.7  | 41.1  | 45.2  | 52.6  | 59.5  | 63.7  | 65.2  | 65.3  | 60.1  | 54.2  | 47.4  | 41.0  |  |  |
|                                                                        | 10%         | DB     | 39.3  | 42.1  | 48.6  | 60.3  | 67.8  | 74.2  | 76.7  | 76.5  | 66.1  | 56.8  | 47.5  | 40.8  |  |  |
|                                                                        |             | MCWB   | 37.2  | 38.4  | 43.0  | 50.5  | 57.3  | 62.1  | 64.0  | 63.4  | 58.2  | 53.0  | 45.3  | 38.8  |  |  |
|                                                                        | 0.4%        | WB     | 45.3  | 46.8  | 50.9  | 59.0  | 64.6  | 70.3  | 70.8  | 70.9  | 65.1  | 59.1  | 52.7  | 46.6  |  |  |
|                                                                        |             | MCDB   | 49.1  | 52.8  | 60.2  | 71.8  | 74.4  | 82.1  | 82.3  | 83.9  | 76.0  | 65.3  | 56.3  | 49.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 42.4  | 43.6  | 48.3  | 56.0  | 62.3  | 67.4  | 68.3  | 68.1  | 62.8  | 56.9  | 49.9  | 43.8  |  |  |
|                                                                        |             | MCDB   | 45.6  | 48.2  | 55.6  | 66.9  | 72.6  | 78.4  | 79.7  | 80.9  | 72.3  | 62.1  | 52.6  | 46.2  |  |  |
|                                                                        | 5%          | WB     | 39.7  | 41.4  | 46.2  | 53.7  | 60.3  | 65.3  | 66.6  | 66.3  | 61.0  | 55.3  | 47.8  | 41.2  |  |  |
|                                                                        |             | MCDB   | 42.5  | 45.6  | 51.8  | 62.3  | 69.2  | 75.3  | 77.6  | 77.5  | 68.5  | 59.1  | 50.1  | 43.3  |  |  |
|                                                                        | 10%         | WB     | 37.3  | 38.7  | 43.8  | 51.4  | 58.3  | 63.4  | 65.0  | 64.5  | 59.2  | 53.4  | 45.4  | 38.8  |  |  |
|                                                                        |             | MCDB   | 39.2  | 42.0  | 48.0  | 59.1  | 65.9  | 71.9  | 74.7  | 74.4  | 64.6  | 56.5  | 47.3  | 40.7  |  |  |
|                                                                        | 5% DB       | MDBR   | 8.9   | 12.1  | 14.5  | 18.0  | 17.8  | 18.7  | 19.0  | 19.4  | 16.4  | 13.2  | 8.8   | 8.5   |  |  |
|                                                                        |             | MCDBR  | 13.0  | 18.2  | 23.9  | 24.4  | 23.4  | 25.1  | 25.8  | 26.4  | 23.2  | 18.6  | 14.7  | 13.3  |  |  |
|                                                                        |             | MCWBR  | 9.8   | 12.0  | 14.5  | 12.9  | 11.3  | 11.0  | 10.5  | 10.9  | 11.6  | 11.3  | 10.4  | 10.3  |  |  |
|                                                                        |             | MCWBR  | 11.8  | 17.0  | 21.3  | 22.3  | 21.1  | 22.2  | 22.3  | 24.1  | 20.3  | 16.8  | 12.6  | 12.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | 9.4   | 11.8  | 13.5  | 12.3  | 10.7  | 10.3  | 9.8   | 10.4  | 10.6  | 10.6  | 9.6   | 10.3  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.293 | 0.315 | 0.355 | 0.394 | 0.397 | 0.399 | 0.409 | 0.404 | 0.376 | 0.347 | 0.321 | 0.292 |  |  |
|                                                                        | taud        |        | 2.390 | 2.365 | 2.312 | 2.250 | 2.280 | 2.304 | 2.277 | 2.307 | 2.355 | 2.414 | 2.440 | 2.440 |  |  |
|                                                                        | Ebn at Noon |        | 244   | 265   | 271   | 270   | 274   | 273   | 268   | 264   | 260   | 249   | 231   | 230   |  |  |
|                                                                        | Edh at Noon |        | 21    | 27    | 34    | 40    | 40    | 40    | 40    | 37    | 32    | 26    | 20    | 18    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 304   | 584   | 913   | 1354  | 1585  | 1728  | 1659  | 1477  | 1051  | 638   | 318   | 236   |  |  |
|                                                                        | RadStd      |        | 51    | 78    | 128   | 172   | 183   | 147   | 166   | 148   | 138   | 80    | 36    | 27    |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | +1.08 | N/A     | N/A    | +1.89   | +1.17 | +0.92 | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (88 neighbors) | +0.92 | N/A     | N/A    | N/A     | +0.68 | +0.61 | N/A         | N/A   | +126  | +54   |

Nomenclature: See separate page